CO4301 – Individual Research Design/ Project

Project Proposal

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Research Project Title	Real time Deep Learning based Segmentation for Complex Medical Images
Key Objectives (Mention appropriately)	1. To develop a deep learning model capable of performing real time segmentation on complex medical images. 2. To apply and evaluate the model using angiogram datasets for accurate region detection. 3. To optimize the model for speed and robustness suitable for real time use. 4. To test generalization on other imaging modalities (MRI, CT, retina images).
Keywords (Maximum 5 separated by ;)	Medical image segmentation; Deep learning; Real time AI; U-Net; Angiogram analysis
Description (Describe the research concisely. Explain why your research is important) Brief about the planned methodology with proper justification. *If necessary give no more than 7 references from reputed sources. *Do not exceed 3 pages	<u>Description</u> Medical image segmentation plays a critical role in assisting clinicians by automatically identifying and separating regions of interest such as vessels, tissues or abnormalities. However, most current segmentation models are either too computationally intensive for real time clinical use or lack accuracy when applied to complex medical images. This research aims to develop a deep learning based segmentation framework capable of performing real time, accurate segmentation of complex medical images. The study will initially focus on angiograms (used in cardiovascular analysis) as the primary application area, while ensuring the method is adaptable to other modalities such as MRI, CT, and retinal imaging. By addressing both accuracy and latency, this project seeks to enhance the reliability of AI assisted medical image analysis. The

	outcome will contribute to the development of fast, generalizable AI systems suitable for clinical workflows and will provide a strong foundation for future research in medical AI and real time deep learning. <u>Brief about the Planned Methodology with Justification</u> Data Collection and Preprocessing: Angiogram datasets from public sources (e.g: DRIVE, STARE or hospital provided data) will be collected. Additional open datasets from MRI or retina images may be used to test model generalization. Preprocessing will include resizing, normalization, and optional denoising to improve consistency and segmentation performance. Model Development: A U-Net based architecture or a lightweight variant (e.g: Mobile U-Net or Attention U-Net) will be implemented and optimized for real time inference. Techniques such as depthwise separable convolutions and model pruning will be explored to improve speed while maintaining accuracy. Training and Evaluation: The model will be trained using annotated datasets with loss functions such as Dice Loss or Binary Cross Entropy. Evaluation metrics will include Dice Coefficient, Intersection over Union (IoU) and inference time per image. Comparisons will be made with standard segmentation models (e.g: classical U-Net, SegNet). Extension and Generalization: The trained model will be tested on other medical imaging modalities to evaluate robustness and transferability. This will demonstrate the method's general applicability across diverse imaging conditions. Justification: The proposed study combines real time performance with generalizable segmentation capability, addressing a major gap in current medical AI systems. The methodology supports future expansion into multimodal medical imaging research, making it directly relevant for postgraduate AI and biomedical engineering applications.

References

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